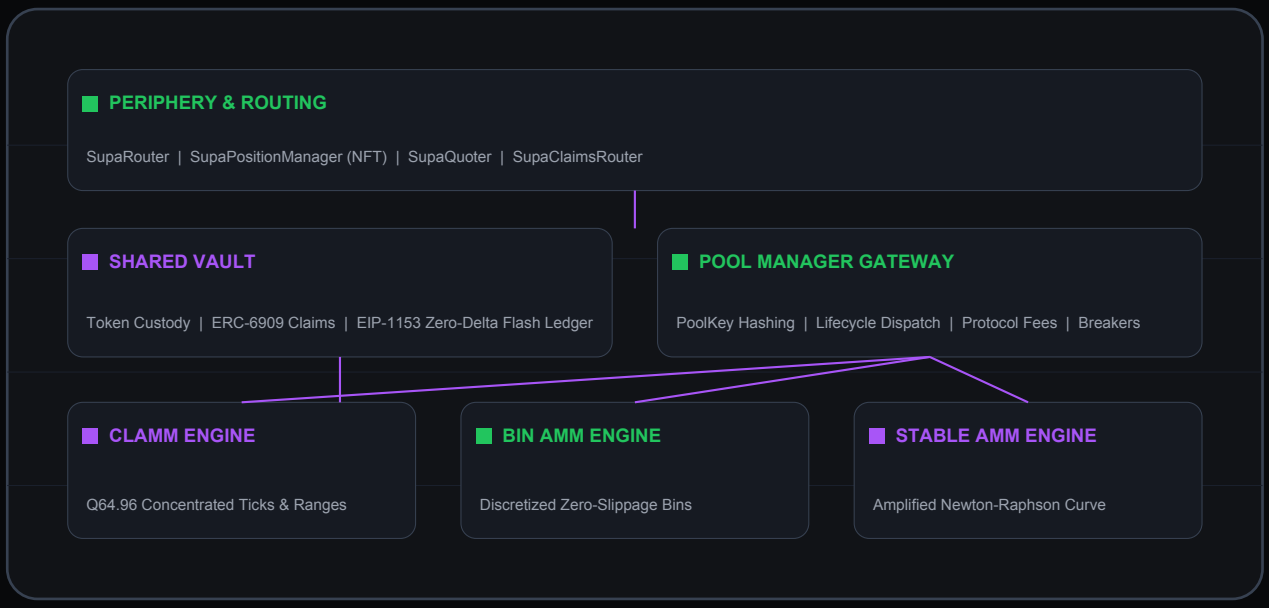
SUPADAO DEX / OFFICIAL PROTOCOL PUBLICATION

Technical Brief

Architecture, evidence, constraints and validation milestones



VERSION 1.0.0 · 15 OCTOBER 2026 · UNDER TECHNICAL REVIEW

Institutional Notice: Repository-derived documentation based on audited smart contract implementations. No independent external audit or public deployment was confirmed.

Document Control & Verification Status

This technical publication is reproducible directly from verified Markdown specifications in the SupaDAO DEX repository. It describes inspected smart contract code and does not constitute legal, tax, or investment advice.

Contents & Architecture Sections

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Proposition

SupaDAO DEX puts pool lifecycle and settlement behind one manager/vault boundary while dispatching specialized liquidity mathematics to three engines. The result is a common integration envelope without pretending that all pool types share one invariant.

Implemented components

- Concentrated liquidity with ranges, ticks, and fee growth.
- Active-bin liquidity with discrete pricing.
- Two-asset amplified stable invariant.
- Transient flash accounting and ERC-6909 claims.
- Router, quoter, position NFTs and protocol fee collection.
- Plugin lifecycle, dynamic-fee experiment, anti-JIT logic, timelock and circuit breakers.

Why it matters

A shared accounting boundary can reduce duplicated custody and routing infrastructure across liquidity models. It also concentrates risk: manager, vault, curve dispatch and governance configuration become critical dependencies.

Evidence and caveats

The repository includes broad Foundry suites and deployment scripts. No independent audit or public deployment was confirmed. The LVR fee signal has a stored-tick limitation, bins do not traverse, and the stable engine does not normalize units. Claims of guaranteed LVR reduction, production security, live TVL or universal best execution are unsupported.

Next validation milestones

Resolve known limitations; reproduce tests and gas snapshots; publish a canonical public-testnet deployment; commission an independent audit; remediate findings; run monitored testnet campaigns; define token-compatibility and operational policies before assessing mainnet readiness.